

Ondřej Kubů

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PROFILE

Research mathematician (ICMAT Severo Ochoa postdoc) moving into AI safety and general-purpose AI (GPAI) risk. Trained in rigorous mathematical reasoning, analysis, and the precise communication of technical results, with 8 peer-reviewed publications in integrable systems and differential geometry. Volunteer with PauseAI CZ, z. s., writing a weekly newsletter that tracks frontier-AI developments and EU AI Act / GPAI governance. Available from late 2026; open to relocation.

AI SAFETY & POLICY ENGAGEMENT

2026 – present

Volunteer, PauseAI CZ, z. s. Author of the weekly newsletter [Pozastavení nad AI](#) (“Pondering AI”). Each issue I monitor primary sources on frontier-AI capabilities, the EU AI Act, and the debate on general-purpose AI risk, judge what matters, and turn it into clear, accurate analysis for a non-specialist audience. Outcome: a consistent publication and a working command of the GPAI risk and governance landscape.

ongoing

Giving What We Can 10% Pledge – committed to donating 10% of lifetime income to effective charities.

EMPLOYMENT

Dec 2024 – Nov 2026

Postdoc (Severo Ochoa), Instituto de Ciencias Matemáticas (ICMAT), CSIC, Madrid, Spain. Research on Haantjes algebras for separation of variables of (super)integrable systems. **Supervisor:** P. Tempesta

Sep 2024 – Nov 2024

Research Associate, Faculty of Nuclear Sciences and Physical Engineering (FNSPE), Czech Technical University in Prague (CTU)

EDUCATION

2020 – 2024

Ph. D., FNSPE CTU, Mathematical Engineering / Mathematical Physics. **Thesis:** *Integrability and superintegrability in the presence of magnetic fields: separable and nonseparable systems*. **Supervisor:** L. Šnobl; **Co-supervisor:** A. Marchesiello. *Rector's Award for Excellent Doctoral Thesis (2025)*.

2018 – 2020

Master's Degree with Honors (Ing.), FNSPE CTU, Mathematical Physics. **Thesis:** *Integrable and superintegrable systems of cylindrical type in magnetic fields*. **Supervisor:** L. Šnobl; **Consultant:** P. Winternitz.

2015 – 2018

Bachelor's Degree with Honors, FNSPE CTU, Mathematical Engineering / Mathematical Physics.

AWARDS

Rector's Award for Excellent Doctoral Thesis, CTU (2025)

Josef Hlávka Award for the best students and graduates of public universities in Prague (2023)

Milan Odehnal Award (2024) – honorable mention (Czech Physical Society)

Dean's Award for the 2nd best Master Thesis (2021)

INTERNATIONAL EXPERIENCE

Spring 2023

ERASMUS+ research exchange, Universidad Complutense de Madrid. **Supervisor:** P. Tempesta

Fall 2019

ERASMUS+ research exchange, Université de Montréal. **Supervisor:** P. Winternitz

TEACHING	2021 – 2024 Assistant lecturer , Dept. of Physics, FNSPE CTU. Seminars in Theoretical Physics 2, Lie Algebras and Groups, and Geometrical Methods in Physics; exercises and consultations for ERASMUS+ students.
SKILLS	Core: rigorous mathematical reasoning and analysis; technical writing and communication for both specialist and general audiences Languages: Czech (native), English (C1–C2), German (B2/C1), French (B2), Spanish (A2) Software: Maple, $\text{\LaTeX}$; author of an open Maple package for Nijenhuis/Haantjes torsion computation (2026)
PUBLICATIONS	8 peer-reviewed articles (all in impacted journals), one further article accepted (in press), and 1 preprint under review; 20 citations (excl. self-citations), h-index 3; 4 peer reviews for <i>J. Phys. A</i>. Full list: ORCID 0000-0001-7463-5680. <i>Preprints</i> O. Kubů, D. Latini. Haantjes algebras, Zernike system and separation of variables, under review. arXiv:2605.23748 (2026) <i>Journal articles</i> O. Kubů, P. Tempesta. Stäckel and Eisenhart geometries, symplectic-Haantjes manifolds and gravitation, accepted, <i>Proc. R. Soc. A</i>; in press. DOI: 10.1098/rspa.2025.1103. arXiv:2509.19950 O. Kubů, D. Reyes, P. Tempesta, G. Tondo. Hamiltonian integrable systems in a magnetic field and symplectic-Haantjes geometry. <i>Proc. R. Soc. A</i> 480 (2024) 20240076 O. Kubů, A. Marchesiello, L. Šnobl. Integrable systems with velocity-dependent potentials: generalized parabolic cylindrical case. <i>J. Phys. A</i> 57 (2024) 235203 O. Kubů, A. Marchesiello, L. Šnobl. New classes of quadratically integrable systems in magnetic fields: the generalized cylindrical and spherical cases. <i>Ann. Phys.</i> 451 (2023) 169264 O. Kubů, L. Šnobl. Cylindrical first-order superintegrability with complex magnetic fields. <i>J. Math. Phys.</i> 64 (2023) 062101 M. F. Hoque, O. Kubů, A. Marchesiello, L. Šnobl. New classes of quadratically integrable systems with velocity dependent potentials: non-subgroup type cases. <i>Eur. Phys. J. Plus</i> 138 (2023) 845 O. Kubů, A. Marchesiello, L. Šnobl. Superintegrability of separable systems with magnetic field: the cylindrical case. <i>J. Phys. A</i> 54 (2021) 425204 S. Bertrand, O. Kubů, L. Šnobl. On superintegrability of 3D axially-symmetric non-subgroup-type systems with magnetic fields. <i>J. Phys. A</i> 54 (2021) 015201 O. Kubů, L. Šnobl. Superintegrability and time-dependent integrals. <i>Archivum Math.</i> 55 (2019) 309–318
SELECTED TALKS	Speaker at international conferences and seminars across Europe and Australia; selected below (full list on website). Invited talk, <i>Geometry and Integrable Systems</i>, La Trobe University, Melbourne, Australia (Feb 2026) Invited talk, <i>Superintegrability, Exact-Solvability, and Representation Theory</i> (Nov 2024) Seminar, Università degli Studi di Milano, Italy (Feb 2026) <i>XXXIII International Fall Workshop on Geometry and Physics</i> (Sep 2025)